

1 Feedback (4 points)

This part is intended to give us some useful feedback on the homework.

1. **Collaboration:** Did you collaborate with anyone on this homework? If so, who? I personally did not collaborate but I shared some of my troubles that I went through with my project partner Hewitt Watkins.
2. **Time:** How long did you spend on this homework? A couple hours every day from Friday - Thursday.
3. **LLM use:** Did you use any LLM? If so, which one(s) and how? Did it help your learning of the material? Yes, I used Gemini to proofread and also to help me make sense of why the autoencoder approach I tried should be abandoned (i was getting extremely low kaggle submission scores).
4. **Comments:** Any comments or suggestions that might help us improve the homework or preparation for it (e.g. anything that was unclear, any readings you like better than the assigned ones)?

2 Paper Exercise: Toward Kernelizing k-Means

2.1

Let z_{ij} be an indicator variable where $z_{ij} = 1$ if $\mathbf{x}_i$ is assigned to cluster j , and 0 otherwise. The total number of points assigned to cluster j is given by $\sum_{l=1}^N z_{lj}$.

So, cluster center is:

$$\mu_j = \frac{1}{\sum_{l=1}^N z_{lj}} \sum_{i=1}^N z_{ij} \mathbf{x}_i$$

Given $\mu_j = \sum_{i=1}^N \alpha_{ij} \mathbf{x}_i$, find coefficients by matching the terms:

$$\alpha_{ij} = \frac{z_{ij}}{\sum_{l=1}^N z_{lj}}$$

2.2

def of squared Euclidean norm in terms of the dot product:

$$\begin{aligned} \|\mathbf{x}_1 - \mathbf{x}_2\|^2 &= (\mathbf{x}_1 - \mathbf{x}_2)^T (\mathbf{x}_1 - \mathbf{x}_2) \\ &= \mathbf{x}_1^T \mathbf{x}_1 - \mathbf{x}_1^T \mathbf{x}_2 - \mathbf{x}_2^T \mathbf{x}_1 + \mathbf{x}_2^T \mathbf{x}_2 \end{aligned}$$

the dot product is commutative ($\mathbf{x}_1^T \mathbf{x}_2 = \mathbf{x}_2^T \mathbf{x}_1$) and if we use this notation $\mathbf{x}^T \mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle$, we get:

$$\|\mathbf{x}_1 - \mathbf{x}_2\|^2 = \langle \mathbf{x}_1, \mathbf{x}_1 \rangle + \langle \mathbf{x}_2, \mathbf{x}_2 \rangle - 2\langle \mathbf{x}_1, \mathbf{x}_2 \rangle$$

2.3

Using part 2, the squared distance from a point $\mathbf{x}_i$ to the center μ_j is:

$$\|\mathbf{x}_i - \mu_j\|^2 = \langle \mathbf{x}_i, \mathbf{x}_i \rangle - 2\langle \mathbf{x}_i, \mu_j \rangle + \langle \mu_j, \mu_j \rangle$$

with the definition of the center we get $\mu_j = \sum_{l=1}^N \alpha_{lj} \mathbf{x}_l$ into the inner product terms:

$$\begin{aligned} \langle \mathbf{x}_i, \mu_j \rangle &= \left\langle \mathbf{x}_i, \sum_{l=1}^N \alpha_{lj} \mathbf{x}_l \right\rangle = \sum_{l=1}^N \alpha_{lj} \langle \mathbf{x}_i, \mathbf{x}_l \rangle \\ \langle \mu_j, \mu_j \rangle &= \left\langle \sum_{l=1}^N \alpha_{lj} \mathbf{x}_l, \sum_{m=1}^N \alpha_{mj} \mathbf{x}_m \right\rangle = \sum_{l=1}^N \sum_{m=1}^N \alpha_{lj} \alpha_{mj} \langle \mathbf{x}_l, \mathbf{x}_m \rangle \end{aligned}$$

Plugging these back in $\langle \mathbf{x}_a, \mathbf{x}_b \rangle$ with kernel notation $K(\mathbf{x}_a, \mathbf{x}_b)$, we get:

$$\|\mathbf{x}_i - \mu_j\|^2 = K(\mathbf{x}_i, \mathbf{x}_i) - 2 \sum_{l=1}^N \alpha_{lj} K(\mathbf{x}_i, \mathbf{x}_l) + \sum_{l=1}^N \sum_{m=1}^N \alpha_{lj} \alpha_{mj} K(\mathbf{x}_l, \mathbf{x}_m)$$